Wireless Networking and Mobile Computing

Emerging Technologies
Audio Communication

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Course Outline

Introduction

1. Data and Signals
2. Data Communication
3. Mobile Systems
4. Mobile Algorithms
5. Mobile AI
6. Cloud Computing

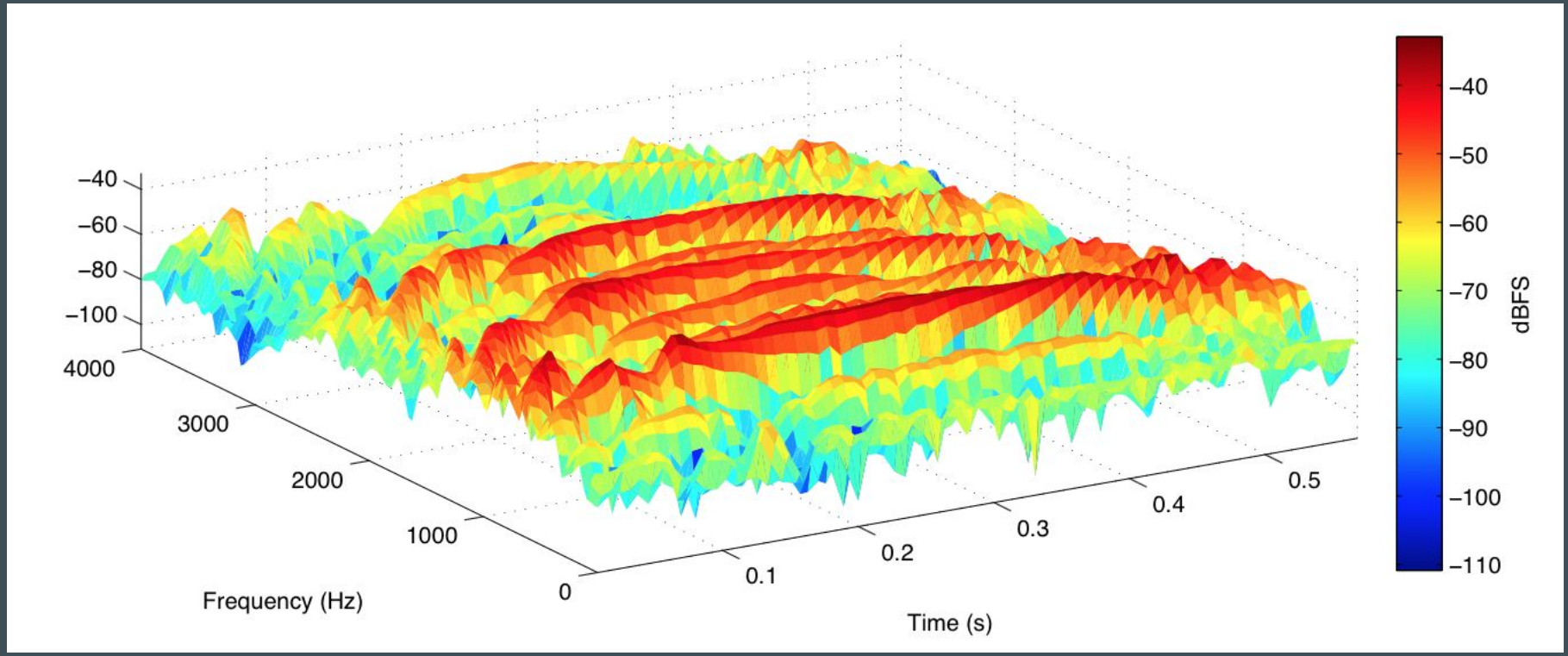
Tournament

Remarks

- Audio sound signals are mechanical waves
  - Sound signals are not electromagnetic signals, not radio signals
- A medium is required to carry the acoustic waves
- Audio signals have a bandwidth, measured in Hertz (Hz)
- A human voice communicates at ~100 Hz ... 1 kHz
  - The human audible frequency ranges up to 15 kHz , and up to 20 kHz for younger adults and children
- Elephants communicate at low frequencies, at ~1 ... 20 Hz, and over a couple of kilometers - Whales: 30 Hz ... 8 kHz

Audio-Spectrogram of the Spoken Word “Hello”

- Level in deciBel (dB) relative to Full Scale (dBFS)



2nd Screen Application

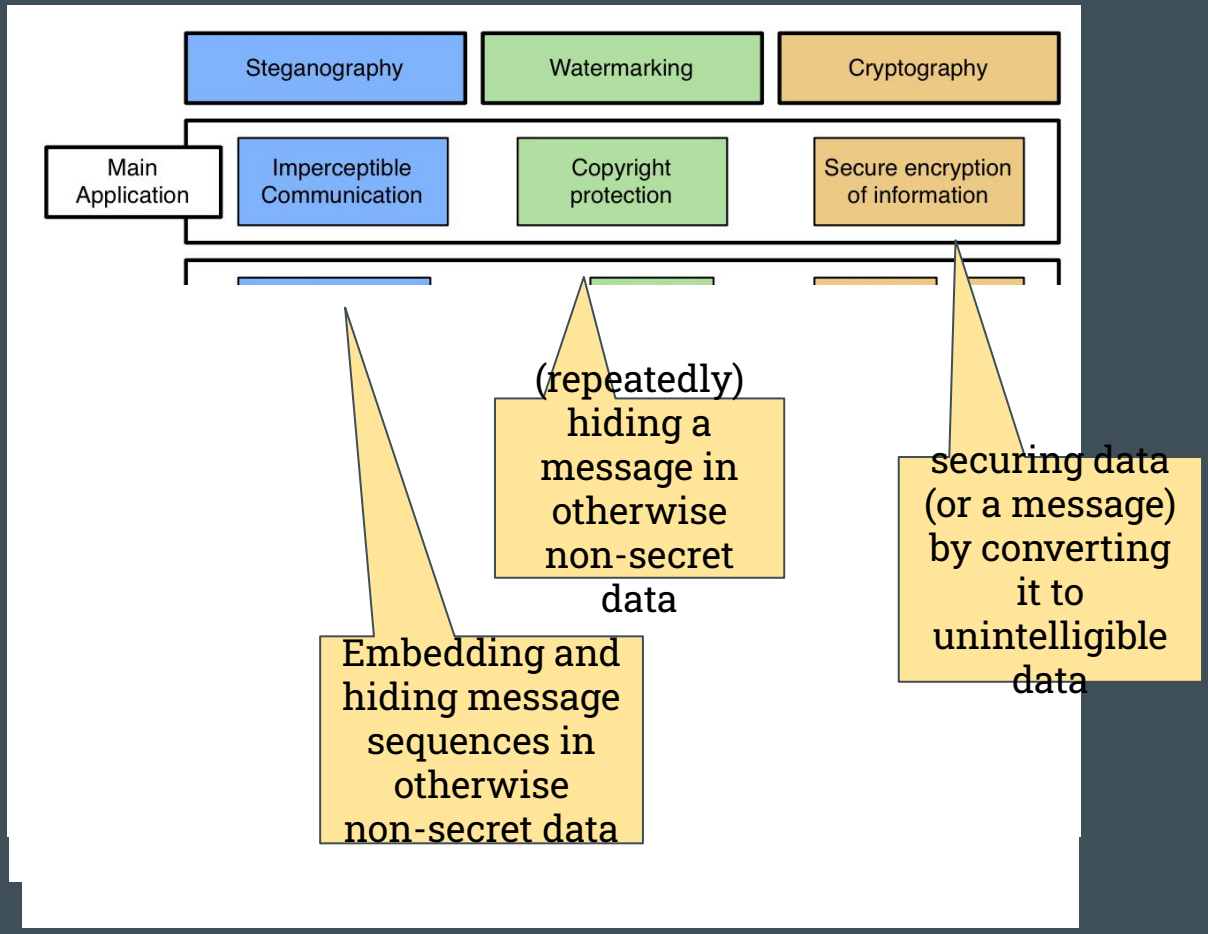


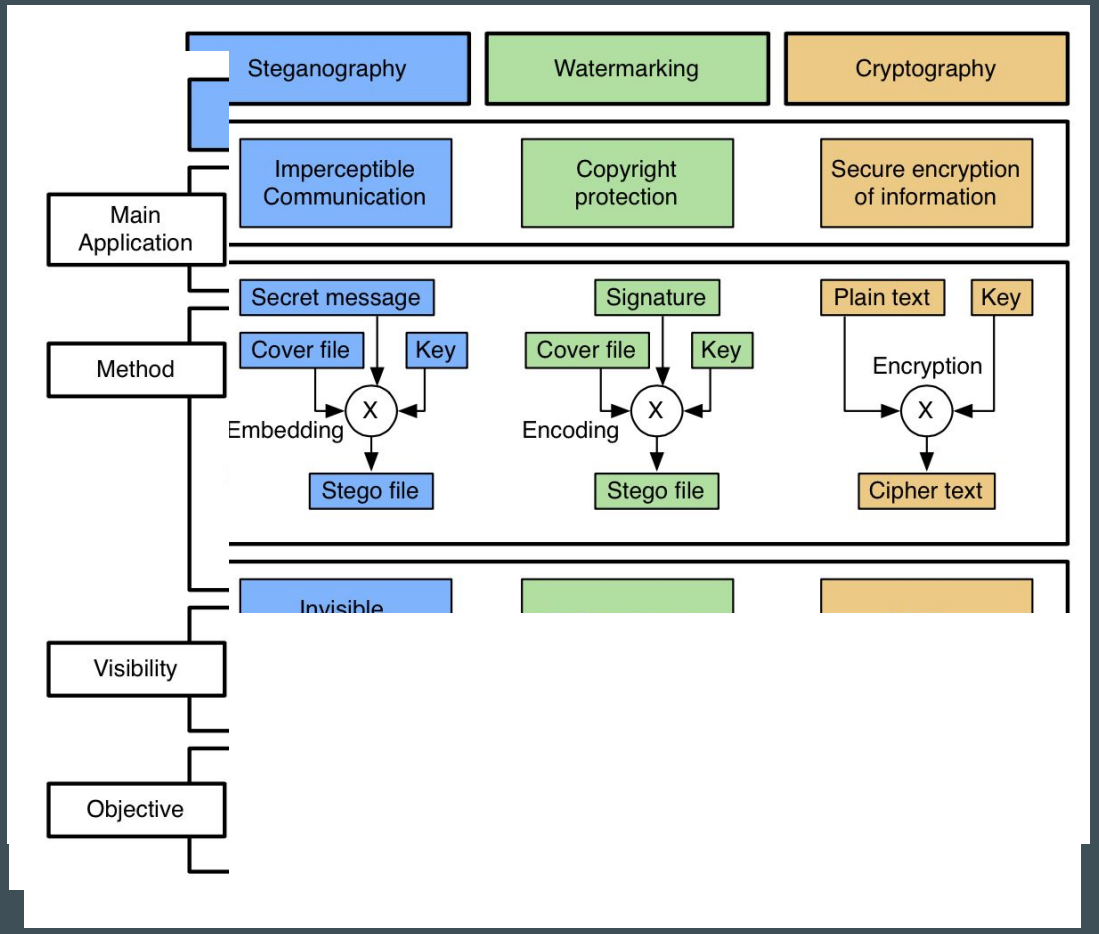
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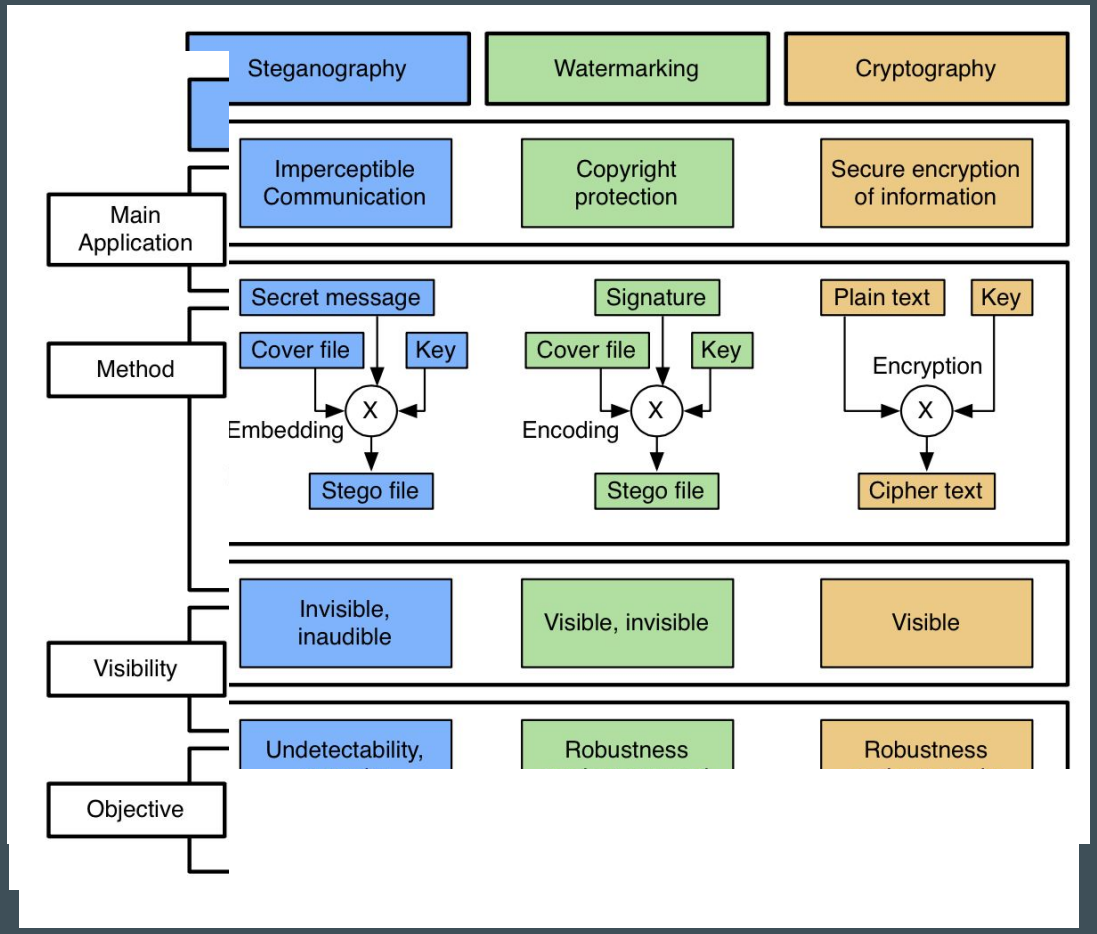


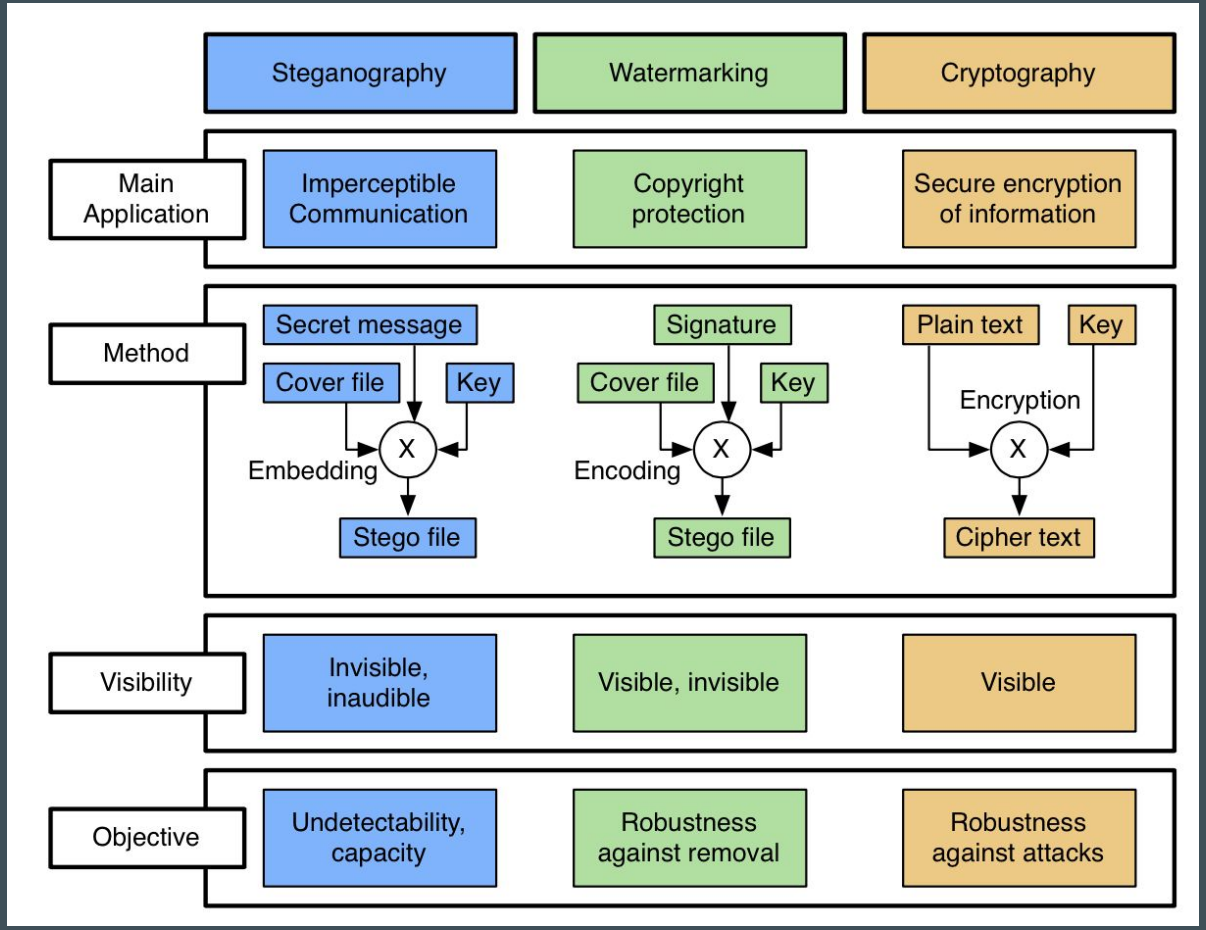
Steganography

- Data Hiding: To avoid detection, hide (secret) information within a non-secret, ordinary medium
- Unlike encryption, which makes data unreadable
- The word "steganography" comes from the Greek words "steganos" (meaning "covered" or "hidden") and "graphein" (meaning "writing")


Steganography
Watermarking
Cryptography
Main Application
Imperceptible Communication
Copyright protection
Secure encryption of information
(repeatedly) hiding a message in otherwise non-secret data
Embedding and hiding message sequences in otherwise non-secret data
securing data (or a message) by converting it to unintelligible data


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Copyright protection
Secure encryption of information
Method
Secret message
Cover file
Key
Embedding
X
Stego file
Signature
Cover file
Key
Encoding
X
Stego file
Plain text
Key
Encryption
X
Cipher text
Invisible
Visibility
Objective


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X
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Key
Encryption
X
Cipher text
Visibility
Invisible, inaudible
Visible, invisible
Visible
Objective
Undetectability,
Robustness
Robustness


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Method
Secret message
Cover file
Key
X
Embedding
Stego file
Signature
Cover file
Key
X
Encoding
Stego file
Plain text
Key
Encryption
X
Cipher text
Visibility
Invisible, inaudible
Visible, invisible
Visible
Objective
Undetectability, capacity
Robustness against removal
Robustness against attacks

Four Methods of Data Hiding

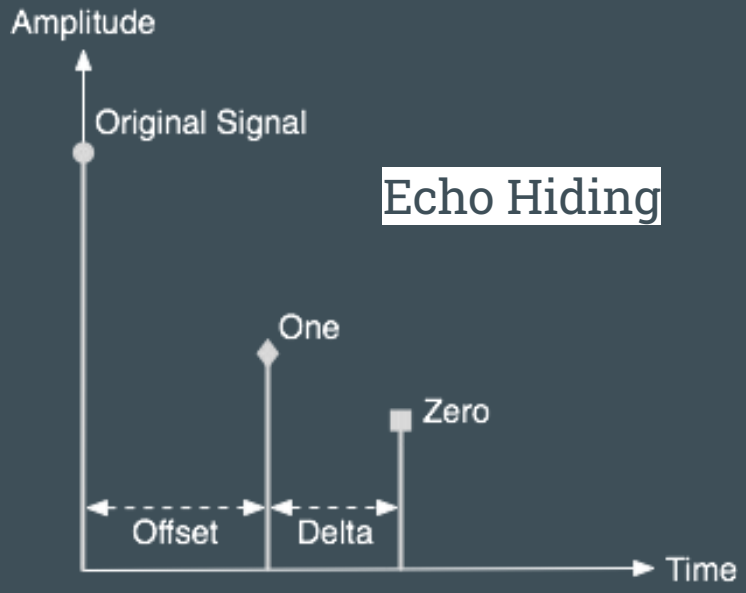
- (1) LSB Coding
- (2) Echo Hiding
- (3) Spread Spectrum Noise
- (4) Phase Coding



Least Significant Bit (LSB) Coding

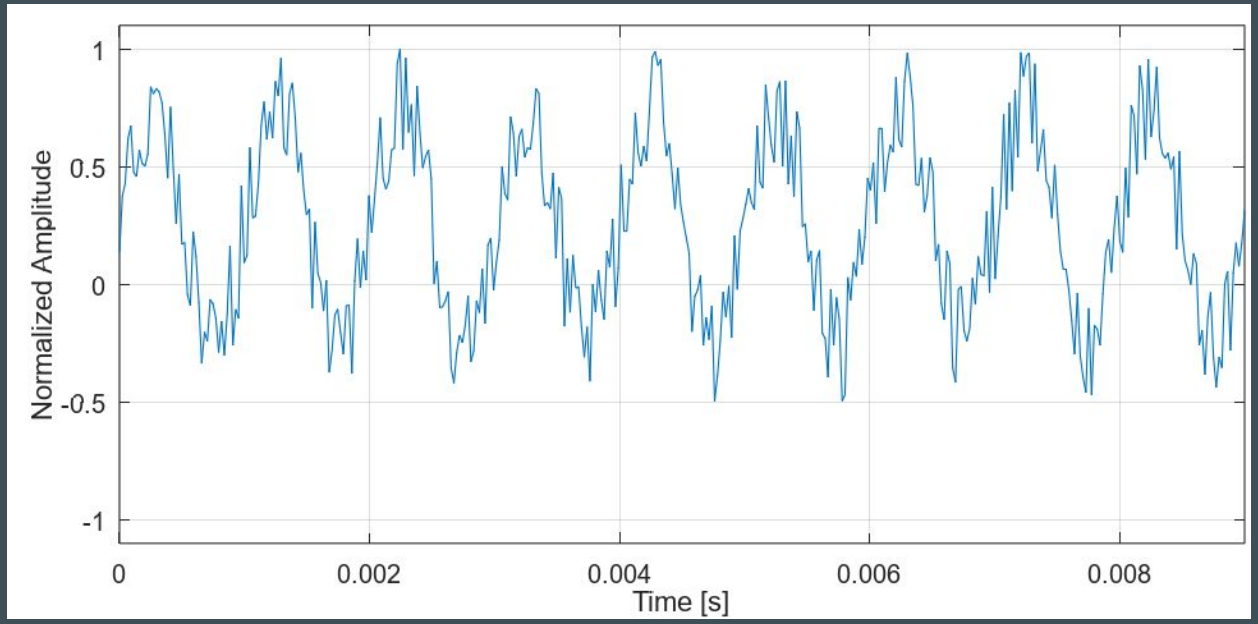
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Four Methods of Data Hiding

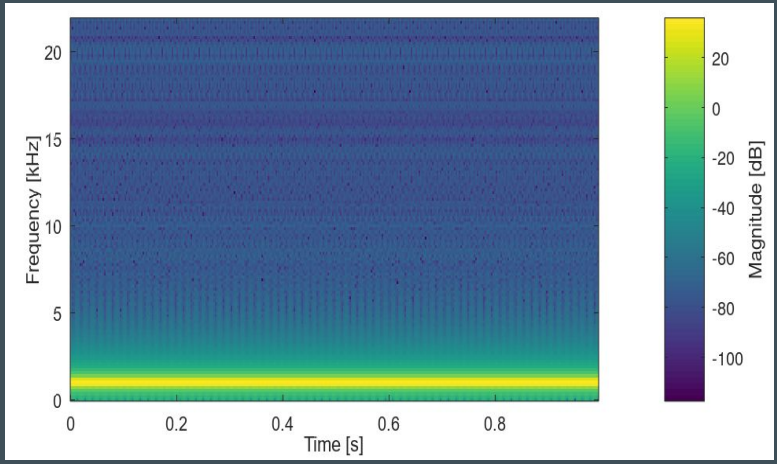
(1) LSB Coding

(2) Echo Hiding

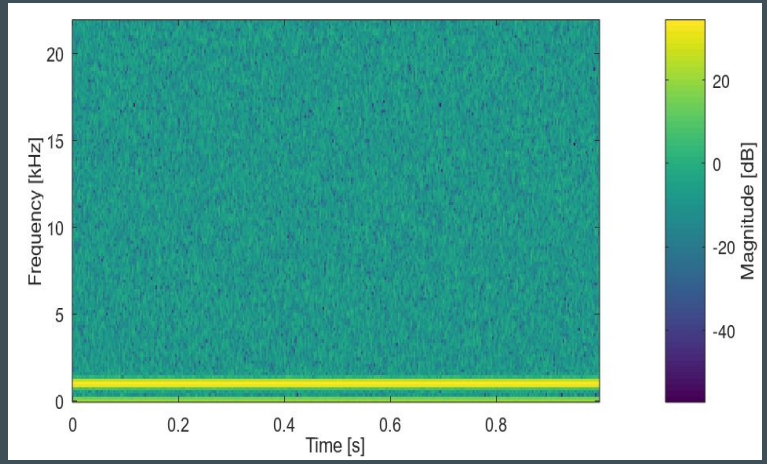
(3) Spread Spectrum Noise

(4) Phase Coding

without noise:

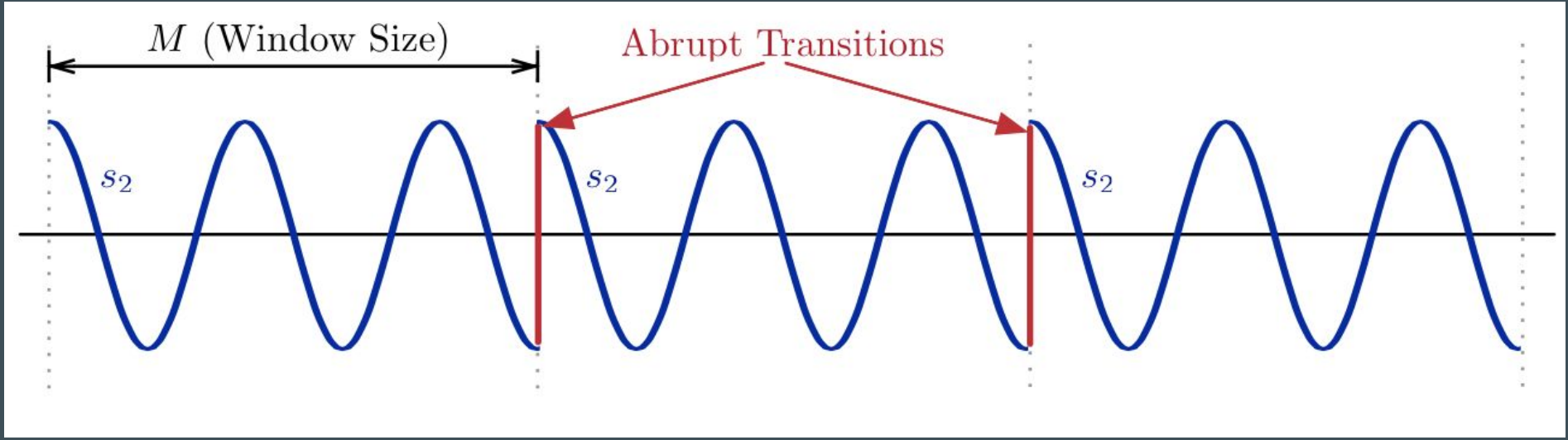


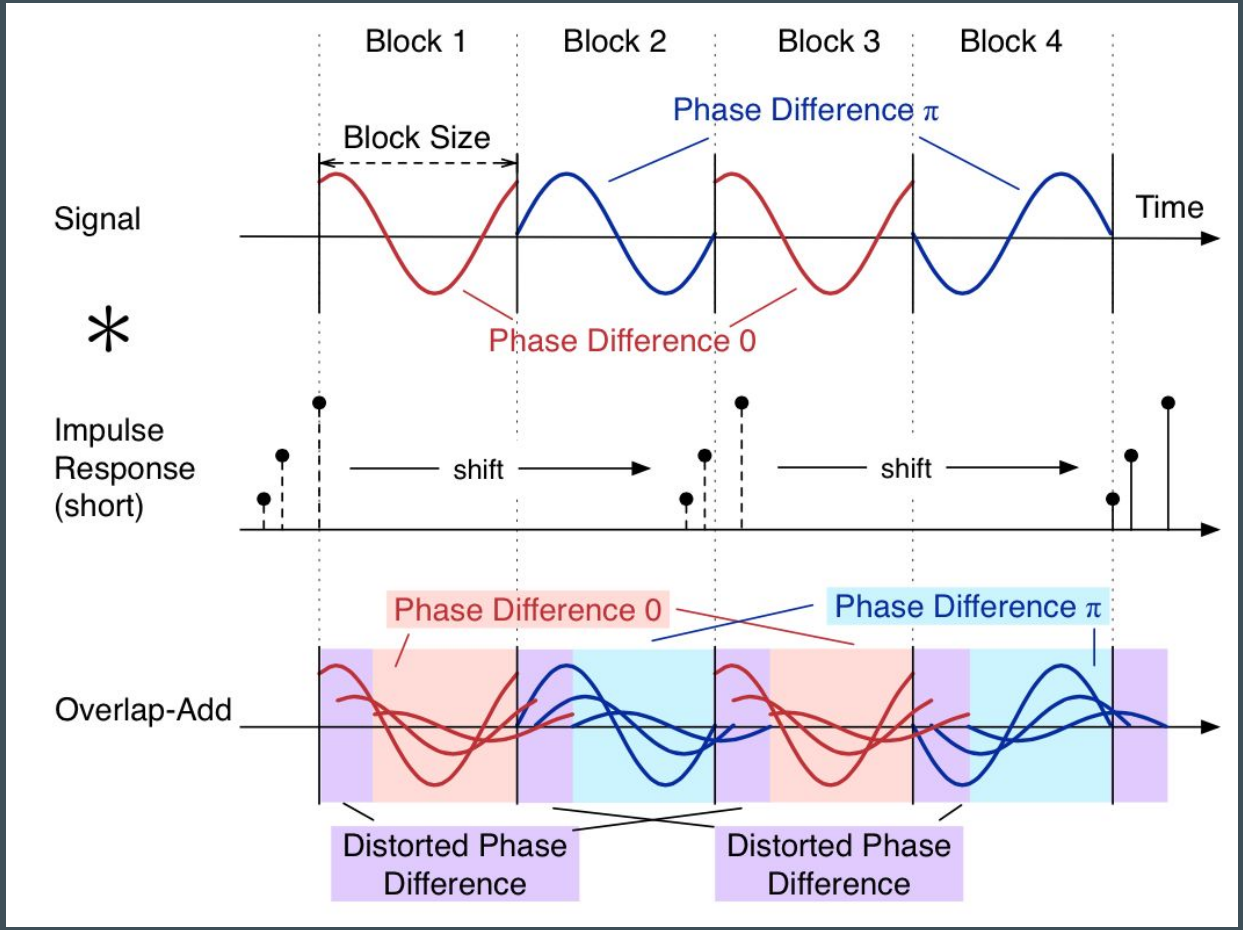
with noise:



Four Methods of Data Hiding

- (1) LSB Coding
- (2) Echo Hiding
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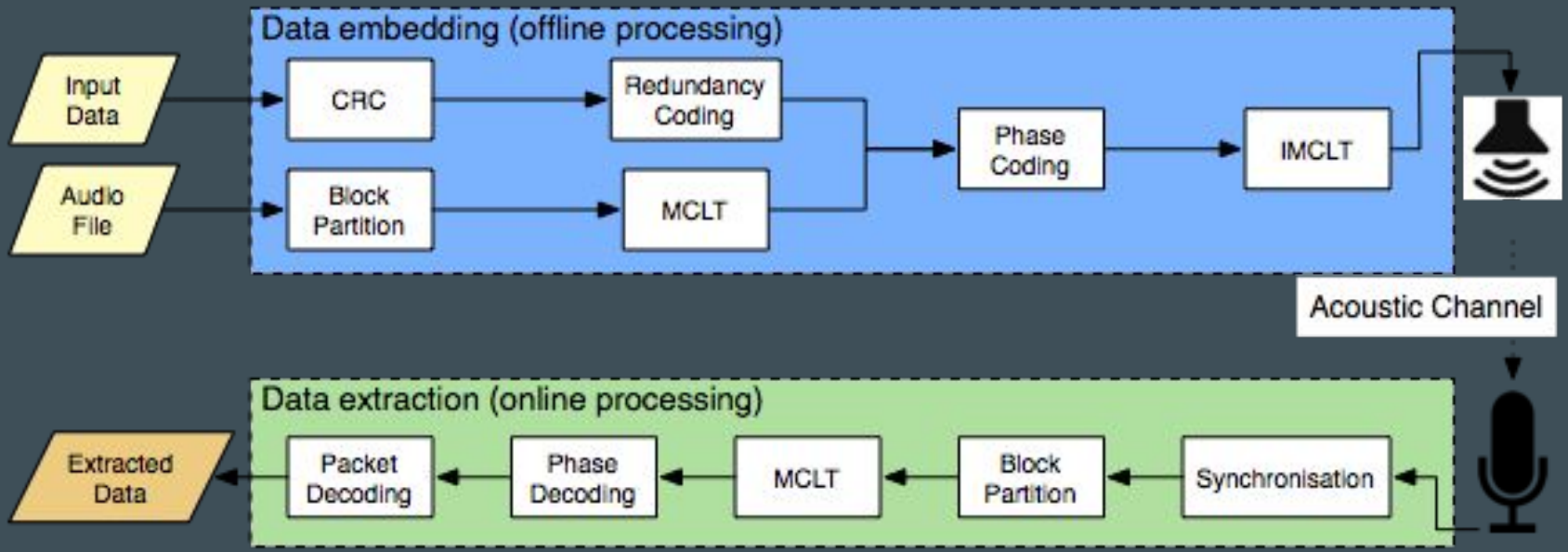

Block 1
Block 2
Block 3
Block 4
Block Size
Phase Difference π
Signal
Time
Phase Difference 0
Impulse Response (short)
shift
shift
Phase Difference 0
Phase Difference π
Overlap-Add
Distorted Phase Difference
Distorted Phase Difference

Phase Coding on Multiple Carrier Frequencies

- Multicarrier phase encoding
- Phase coding has advantages over alternative solutions
  - Shazam fingerprinting: only fingerprinting, no data communication, requires training
  - Ultrasound: increased audio signal energy (not useful for TV channels)
  - LSB Coding, Echo hiding, Spread Spectrum Noise: also increased audio signal energy, plus additional background noise

System Overview

- Modulated Complex Lapped Transform (MCLT)
  - Creates multicarrier channels
  - Similar to Fast Fourier Transform (FFT)



MCLT and Fourier Transform

- Discrete Fourier Transform
  - k denotes the frequency

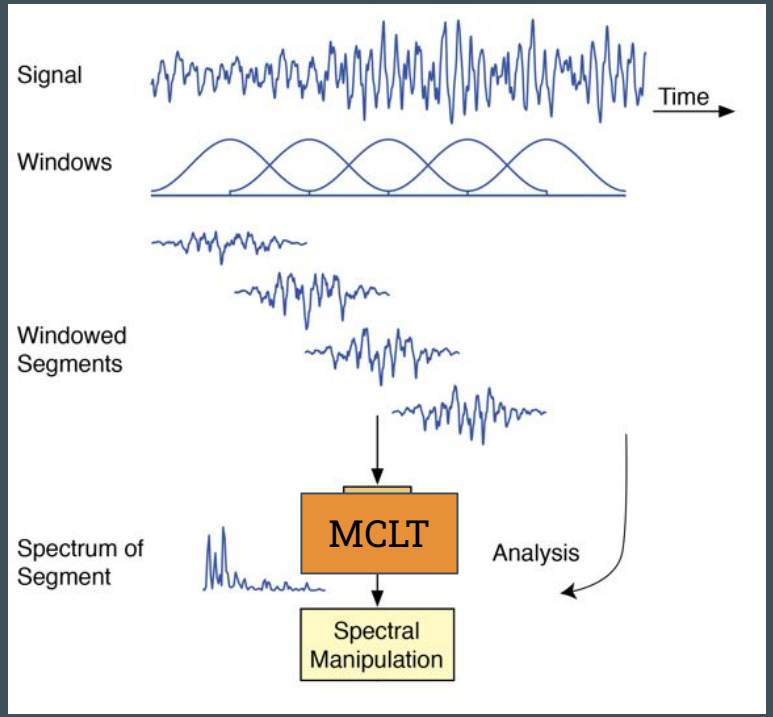
$$X[k] = \sum_{n=0}^{N-1} x[n] \cdot e^{-j2\pi k \frac{n}{N}}$$

- MCLT: Fourier with extra window (“lap”) and phase correction
  - Data is hidden in the phase
  - “M” denotes the frame, aka the time index

$$X_m[k] = \sum_{n=0}^{N-1} x[n] \cdot w[n - m\frac{N}{2}] \cdot e^{-j\left(2\pi k \frac{n}{N} + \phi_k\right)}$$

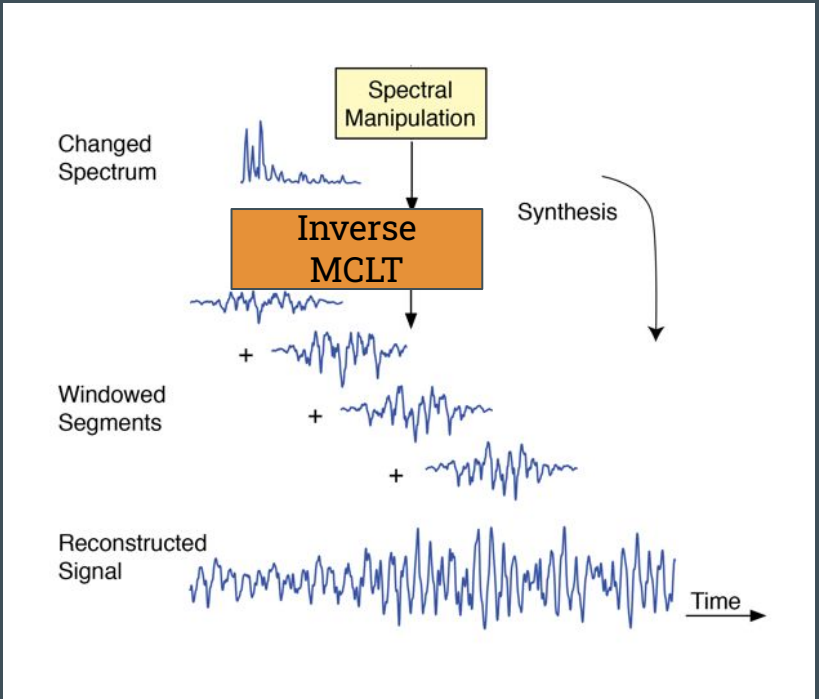
Signal Construction - Transmission

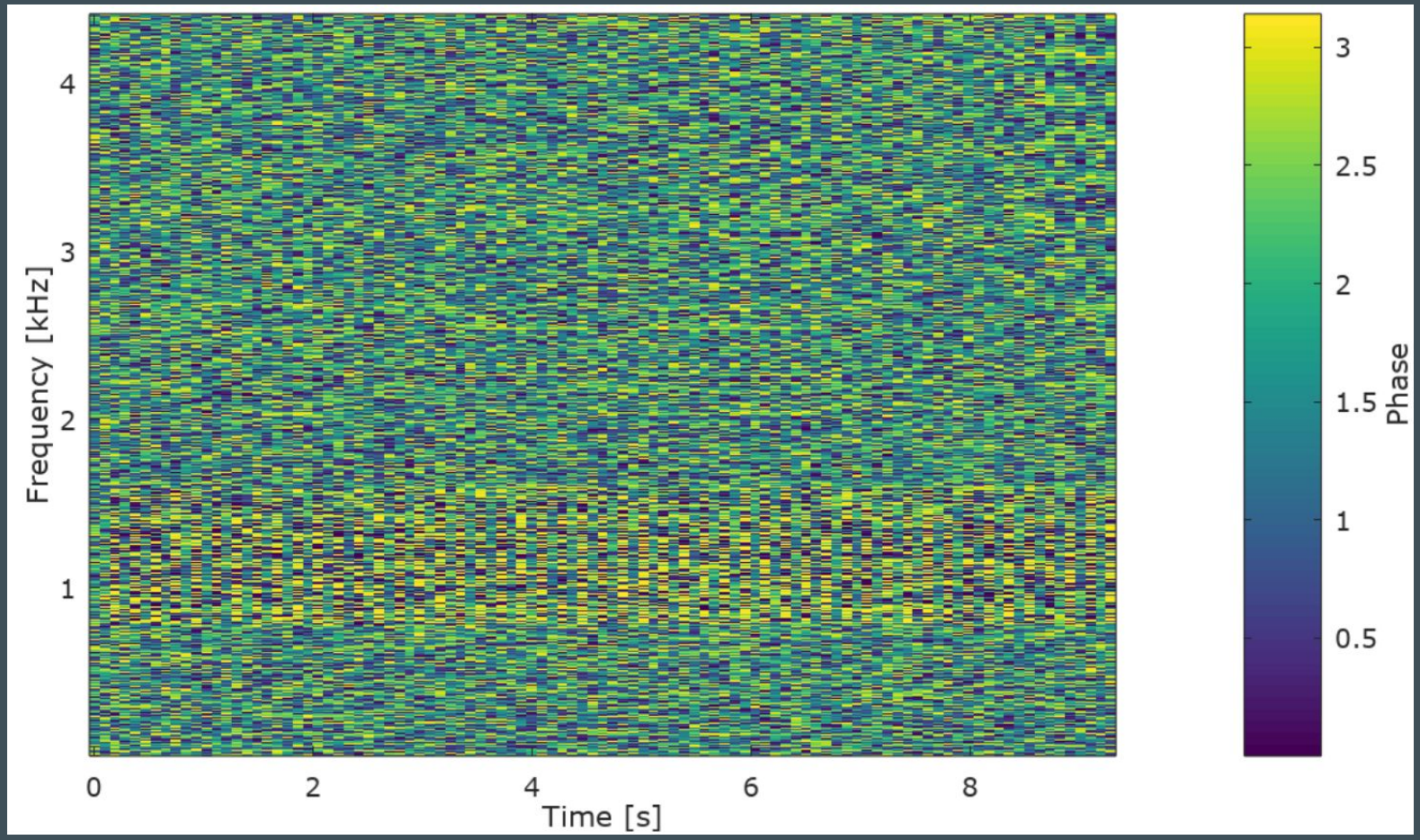
- Offline construction, or immediately in real-time



Signal Reconstruction - Reception

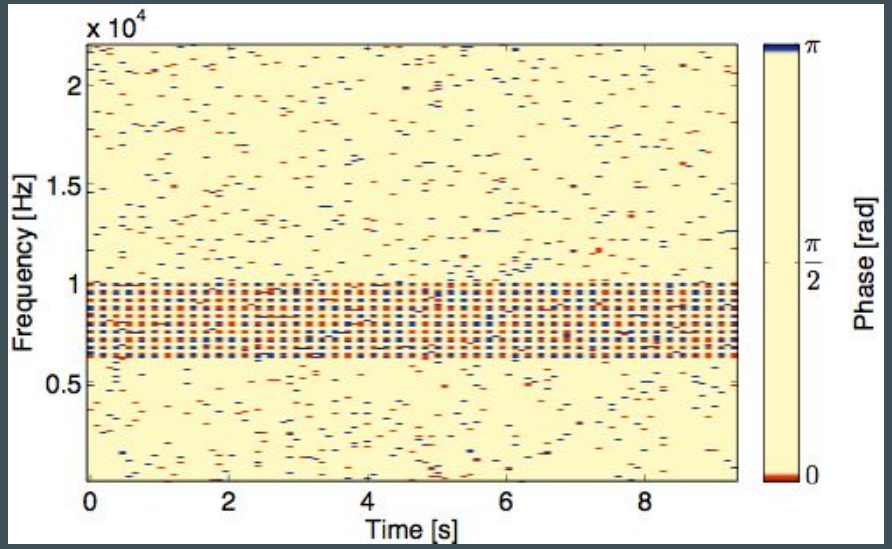
- Offline construction, or immediately in real-time



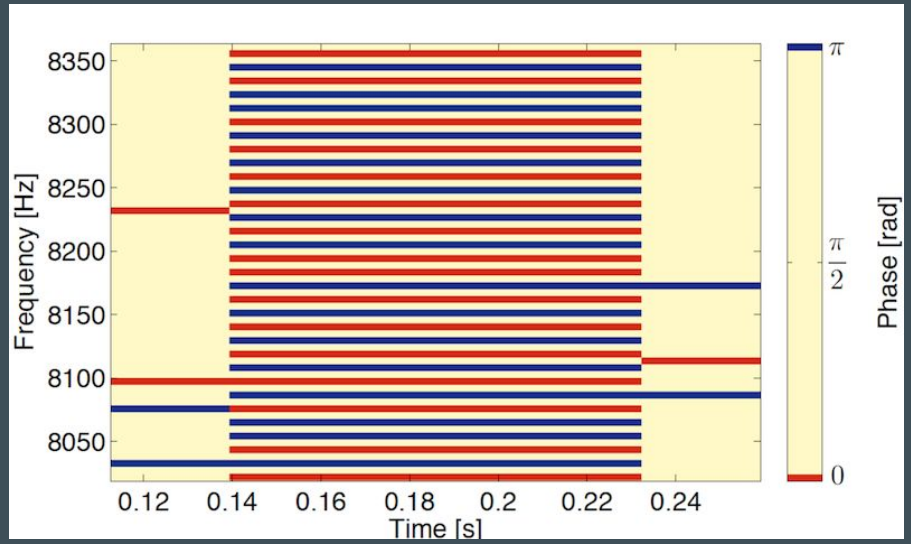

Frequency [kHz]
4
3
2
1
0
2
4
6
8
Time [s]
Phase
3
2.5
2
1.5
1
0.5

Resulting Phases after Signal Construction

- Data is hidden in phase information
- Signal Reconstruction = demodulating the phase

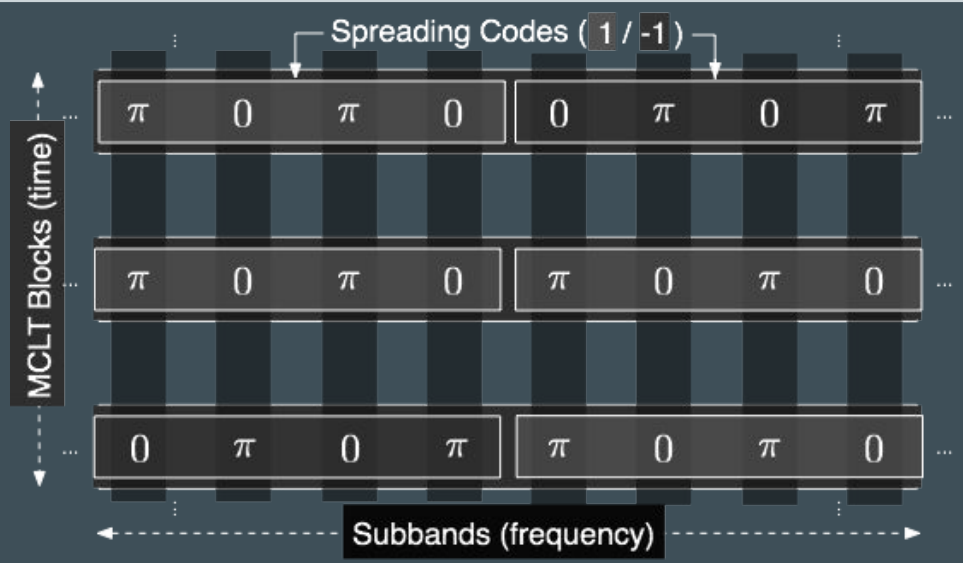


encoded data



zoomed

Phase Coding Spread over Four Subbands

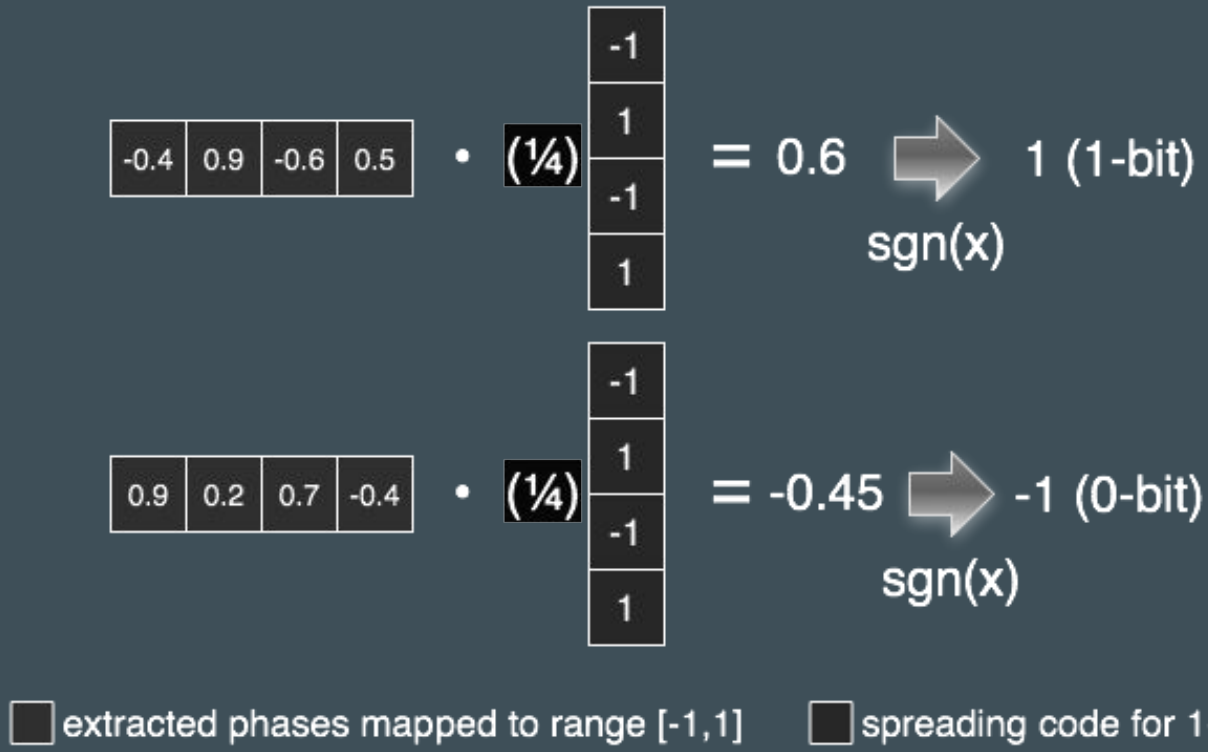


- Bits translated to spreading codes:

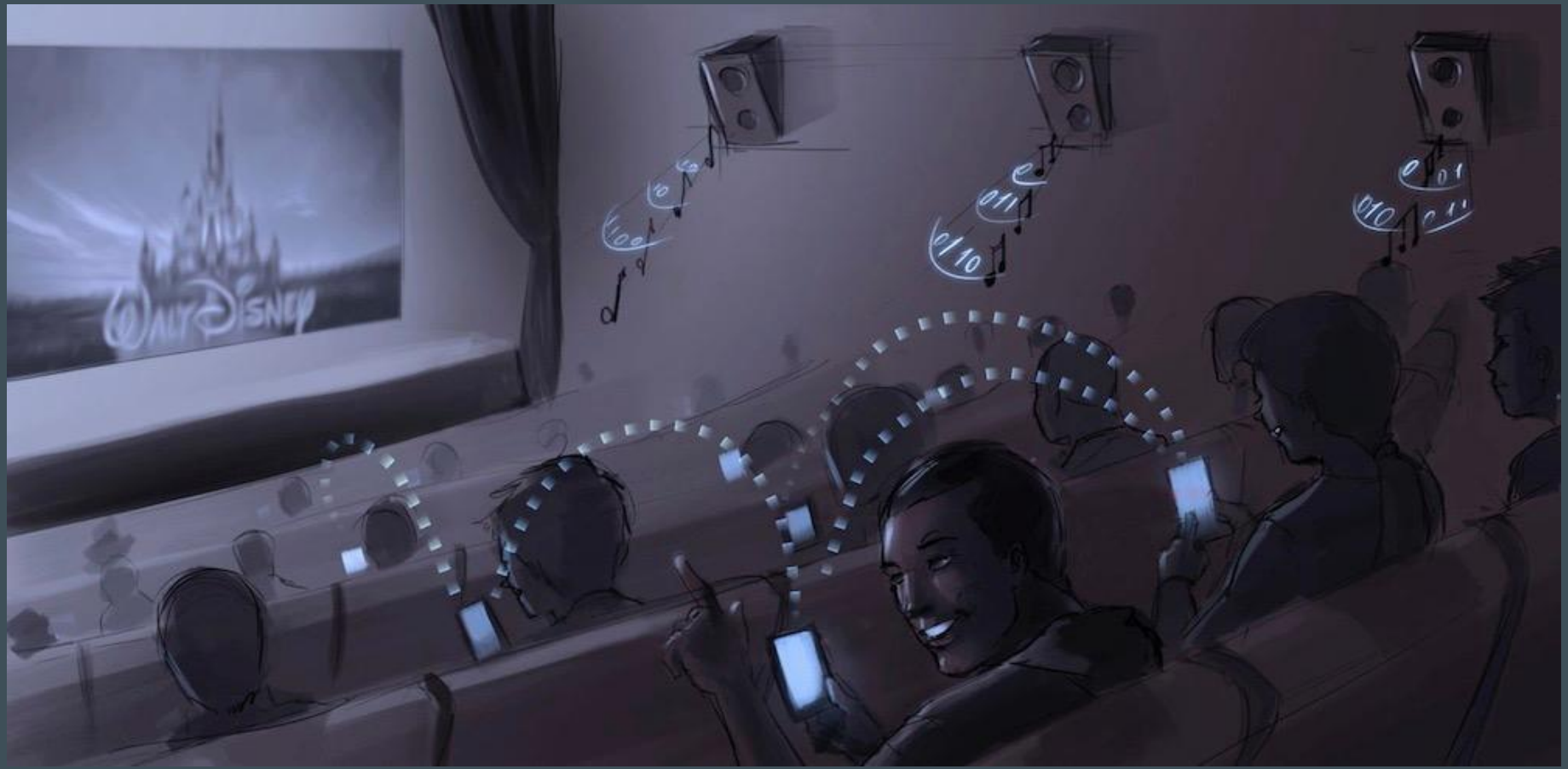
$$1 \rightarrow [-1\ 1\ -1\ 1] \rightarrow [\pi\ 0\ \pi\ 0]$$

$$-1 \rightarrow [1\ -1\ 1\ -1] \rightarrow [0\ \pi\ 0\ \pi]$$

Phase Decoding (Correlation)

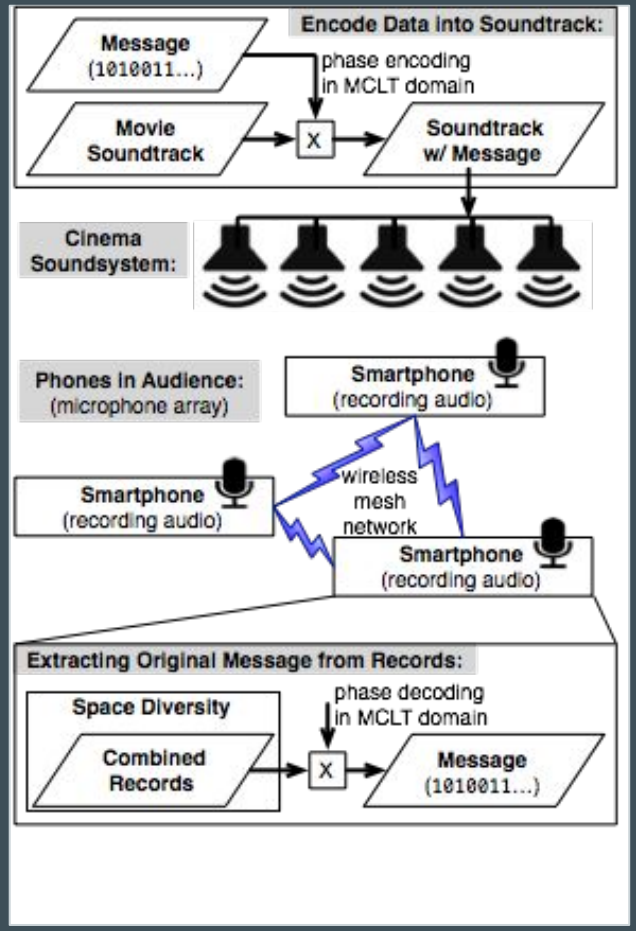


Receiver Diversity (Use of Multiple Phones)



Prototype

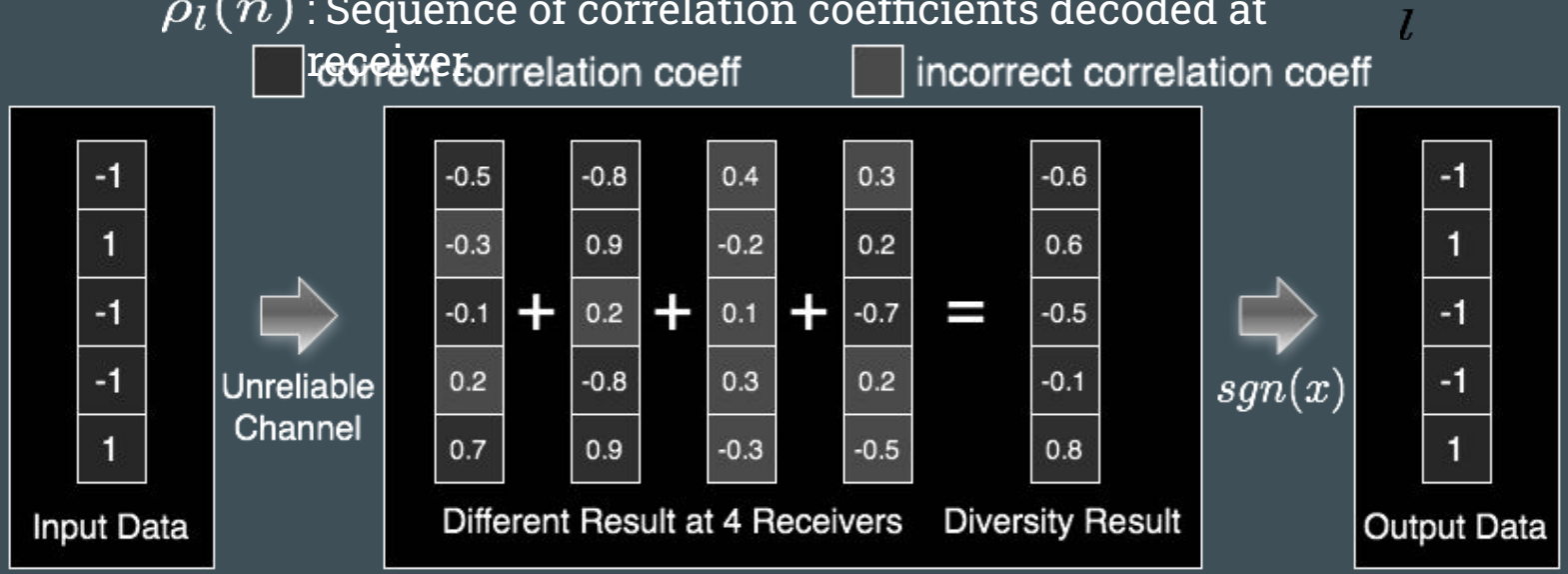
- Embedding of messages
  - “Audiocom”
  - iOS app as receiver/extractor
- Personal Hotspot for ad-hoc network (Wi-Fi)
- Single server combines all receivers’ sequences



Receiver Diversity

$$\hat{\rho}(n) = \frac{1}{L} \sum_{l=0}^{L-1} \rho_l(n)$$

$\rho_l(n)$: Sequence of correlation coefficients decoded at receiver l




ETH
Disney Research

Tests in Cinema

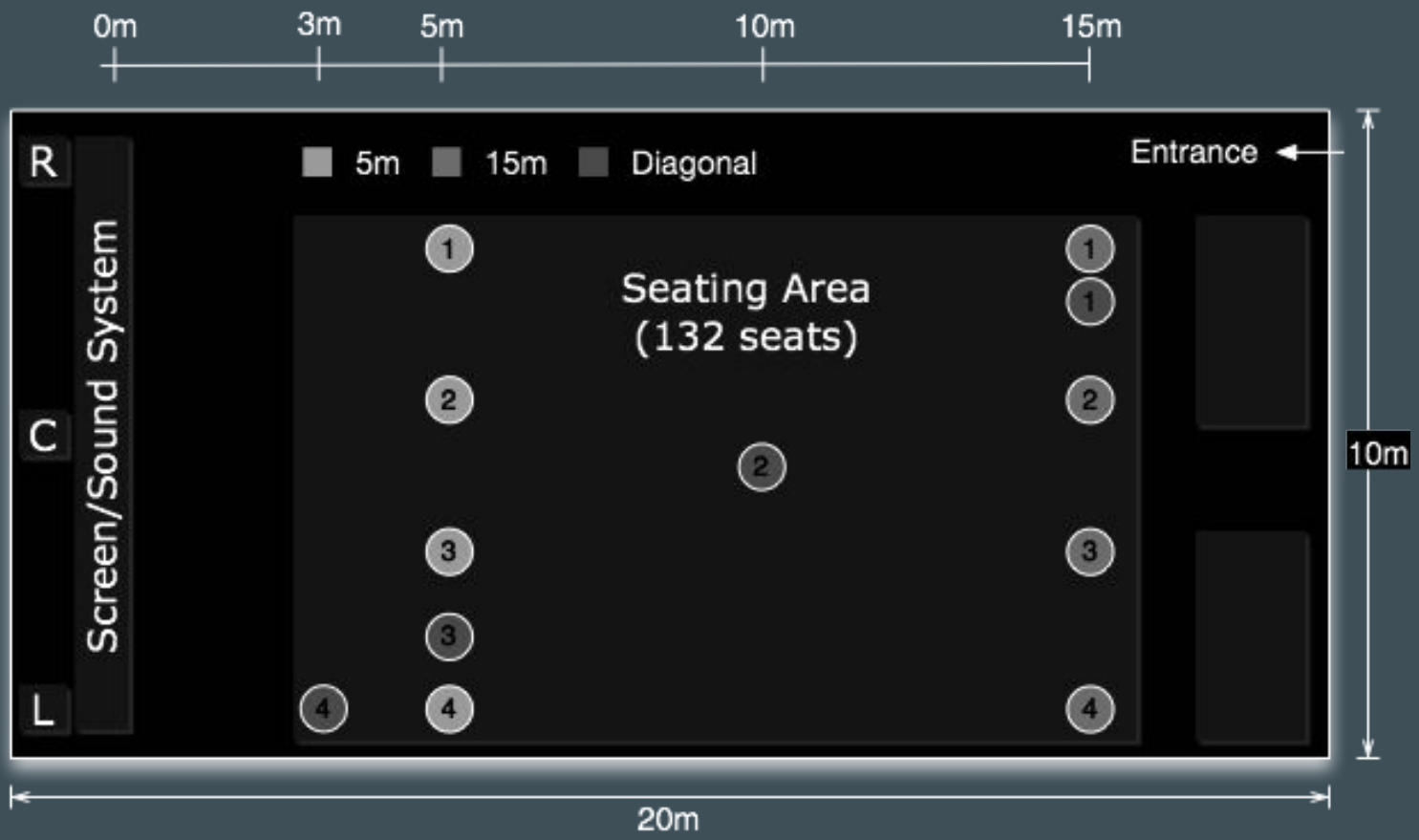
- Arena Cinemas Zurich
- 200 square meter cinema (10m x 20m)
- Left, center, right speakers
- 4 receivers
  - Carrier signal: Popular Music
- Metric:
  - Bit Error Rate (BER)



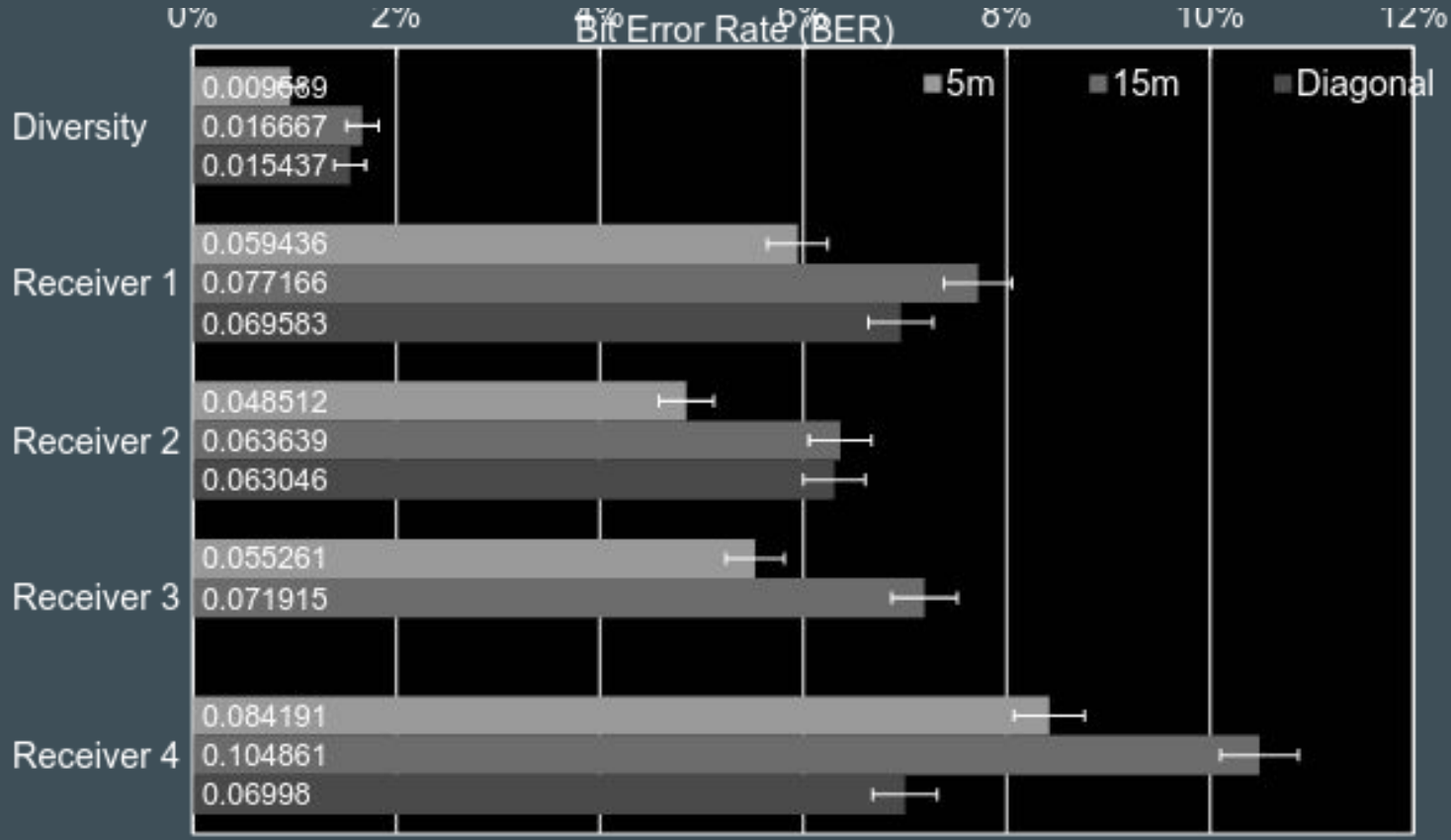

CINEMAS



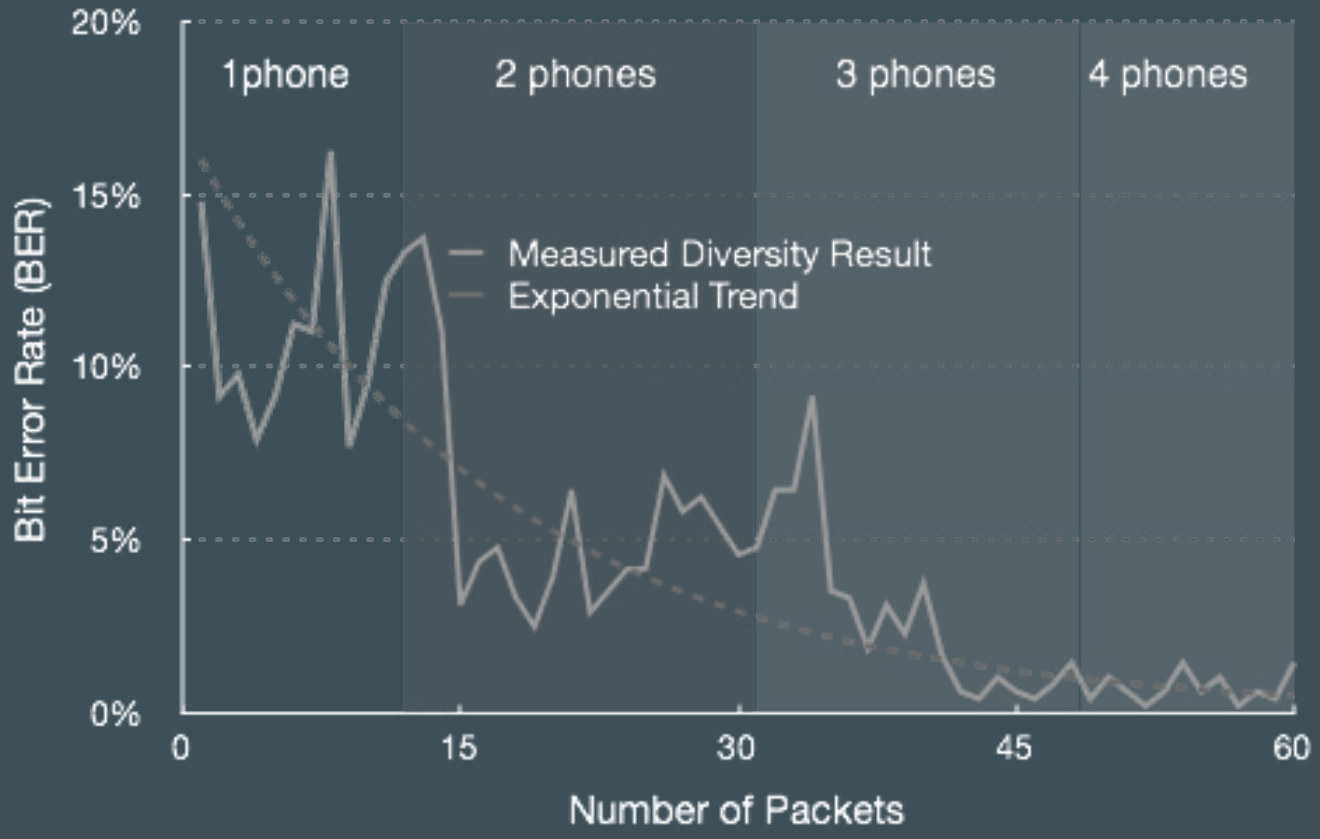
Cinema Field Test: Receiver Arrangement



Cinema Field Test: Results (1)



Cinema Field Test: Results (2)



Exercise: Acoustic Communication

- Open in your course folder:
  02_Notebooks/02_Data_Communication/
  02.10_Acoustic_Communication.ipynb
- Compare ASK, FSK, and PSK; inspect secret_message.wav.
- Complete Section 10.8 on modulation, rate, and sampling.
- Save your responses and submit via Moodle
  as instructed in the assignment guidelines.

Exercise: Acoustic Noise and Detection

- Open in your course folder:
  02_Notebooks/02_Data_Communication/
  02.11_Acoustic_Noise_and_Detection.ipynb
- Inspect noise, vary SNR, and compare FSK detection.
- Complete Section 11.7; identify the secret-message tones.
- Save your responses and submit via Moodle
  as instructed in the assignment guidelines.

Exercise: Synchronization and Framing

- Open in your course folder:
  02_Notebooks/02_Data_Communication/
  02.12_Acoustic_Synchronization_and_Framing.ipynb
- Detect the chirps and recover the secret-message frame.
- Complete Section 12.8; explain timing and decoding evidence.
- Save your responses and submit via Moodle
  as instructed in the assignment guidelines.

Exercise: Over the Air Communication

- Open in your course folder:
  02_Notebooks/02_Data_Communication/
  02.13_Over_the_Air_Communication.ipynb
- Work in pairs to exchange messages between two devices.
- Complete Section 13.8; record settings, results, and failures.
- Save your responses and submit via Moodle
  as instructed in the assignment guidelines.

